Electronic Principles and Devices · Unit_01 · Unit-1 Semiconductor Diode

# DRAFT — UE26EC141A · Unit_01 · Unit-1 Semiconductor Diode · Ideal versus Practical diode, Diode resistances

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

# · How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

| | |
|---|---|
| Source text | Verbatim or OCR-derived text, kept separate from editorial notes. |
| Candidate | Automated topic/source association requiring manual confirmation. |
| Low ASR confidence | Transcript segment requires audio verification. |
| DRAFT | Human review remains incomplete. |

## Contents

1. 1 · Ideal versus Practical diode, Diode resistances
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

# Ideal versus Practical diode, Diode resistances

## 1 Ideal versus Practical diode, Diode resistances

Study focus: Ideal versus Practical diode, Diode resistances. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000038 · EV0000039

SHORT-NOTE RECAP

**Core explanation · 00:00–04:59**

I'm Professor, I'm in the Department of Electronics and Communication Engineering.

**Key details**

- Today, we're going to learn the ideal and practical diode characteristics along with the diode resistances.
- As we see from the diagram, we're going to learn about the equivalent representation of the diode, this has been written based on the characteristics curve.
- There are two types of practical diodes out there.
- There are two types of diodes representations are there.
- One is ideal diode.
- When an ideal diode is forward biased, as shown in the diagram here, it offers no resistance and act like a closed switch.
- Likewise, the ideal diode under reverse bias offers infinite resistance, hence it is a open switch.
- Coming to the practical diode as shown, the diode which is said to be forward biased, it starts conducting at a new voltage, which is 0.7 for silicon and reverse bias, no current, and this practical diode is considered as, again, the open switch, similar to that of an ideal diode.
- For germanium, it would be 0.3.
- So the next point, what we are going to understand is diode resistances.
- An actual diode offers a very small resistance, non-zero, when forward biased, and is called the forward resistance.
- Whereas it offers very high resistance, not infinite, when reverse biased, and is also called as a reverse resistance.
- Type of applied voltage or a signal will define the following resistance levels.

- Coming to the first DC or a static resistance, the application of an DC voltage to a circuit containing a semiconductor diode will result in operating point on a semiconductor diode characteristics curve, which is x-axis is VD, y-axis is.
- Typically, the DC resistance of a diode is for forward bias, 10 ohm to 80 ohm.
- Reverse bias, 10 megaohm.
- Coming to the AC or a dynamic resistances, the application of a small AC voltage to a circuit containing a semiconductor diode offers AC resistance.
- A straight line drawn tangent to the curve through Q point will define a particular change in voltage and a current.
- That change in the voltage and change in the current determine the AC or a dynamic resistance, which is shown that is RD is equal to delta VD by ID for this region of the diode characteristics.
- Coming to AC or a dynamic resistance, dynamic resistance can be found simply by substituting the quotient value of the diode current into the equation as RD is equal to 26 millivolts by ID, or RD dash is equal to 26 millivolts by ID plus.
- This RB is the bulk current body resistances.
- Typically, the AC resistance of diode in active region will range from 1 ohm to 100 ohm.
- Further, the average AC resistance can also be calculated with a diagrammatic from the diode characteristics as shown.
- If the applied input voltage is sufficiently large to produce a broad swing and a resistance determined by the straight line drawn between the two intersection established by a maximum and a minimum value of the input voltage is called the average AC resistance.
- That is, RAV is equal to delta VD by delta V at point to point.
- Then average AC, defined by the straight line between the limits of the operation.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

**Core explanation · 05:00–09:59**

Determine the AC resistance ID is equal to 25 milli.

**Key details**

- Then compare the results of and to the DC resistance at the each level.
- 2 milli is the center point.
- Then substitute the value, we get 27.5.
- And substitute here, we get 2 milli.
- Coming to the point 3, compare the results of and to the.
- For a DC resistance at 2 milli, we have 0.7, we get 30, 50, RD is equal to 31.
- This is because in AC, we use a small value change, whereas in DC, we don't take a small value change, we take a large value change.
- Before we go for second numerical, there could be one derivation of expression for.
- AC or a dynamic resistance.
- We know that RD is equal to 26 millivolts by ID.
- How to derive this?

- KT ID plus IS.
- Then arrange, whereas we know that this is where IS reverse saturation, very less current.
- We know that K is equal to 11,600, TK is 298.
- After substitution, we get RD is equal to 26 millivolts by ID.
- This could be asked in the exam for some formulas.
- And the fifth numerical, here first they're asking, determine the DC resistance levels.
- In the previous numerical, we solved for AC, now for the DC at 2 milli, at 20 milli, and -10 milli, these are the reverse point.
- After substitution, we get the different values over there.
- In this comparison, for forward bias, the resistance is very less.
- Reverse bias, the resistance is very high, around 10 megaohms.

**Example or application from the lecture**

- Then coming to question number three, calculate the DC and AC resistance of the diode shown in the figure given below at a forward current of 10 milli, and also compare their magnitudes.

*These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.*

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

# A Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|

# B Sources, coverage and gaps

| Source | Path | Use |
|---|---|---|
| Ideal versus Practical diode, Diode resistances transcript (JSON) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/005_Ideal_versus_Practical_diode_Diode_resistances__3b08a07358669379.transcript.json | Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |
| Ideal versus Practical diode, Diode resistances transcript (Markdown) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/005_Ideal_versus_Practical_diode_Diode_resistances__3b08a07358669379.transcript.md | Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

| Source | Path | Use |
|---|---|---|
| Ideal versus Practical diode, Diode resistances transcript (plain text) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/005_Ideal_versus_Practical_diode_Diode_resistances__3b08a07358669379.transcript.txt | Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps. |
| Ideal versus Practical diode, Diode resistances transcript (SRT) | https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/005_Ideal_versus_Practical_diode_Diode_resistances__3b08a07358669379.srt | Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps. |

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

## Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

## Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

| Source | Unit counts | Example decisions |
| --- | --- | --- |
| SRC00018 | 2 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0111. Sidecar variants compare... |
| SRC00019 | 2 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0111. Sidecar variants compare... |
| SRC00020 | 2 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0111. Sidecar variants compare... |

| Source | Unit counts | Example decisions |
| --- | --- | --- |
| SRC00021 | 2 covered · 0 excluded · 0 gaps | t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare...<br>t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0111. Sidecar variants compare... |

# C Revision and retrieval practice